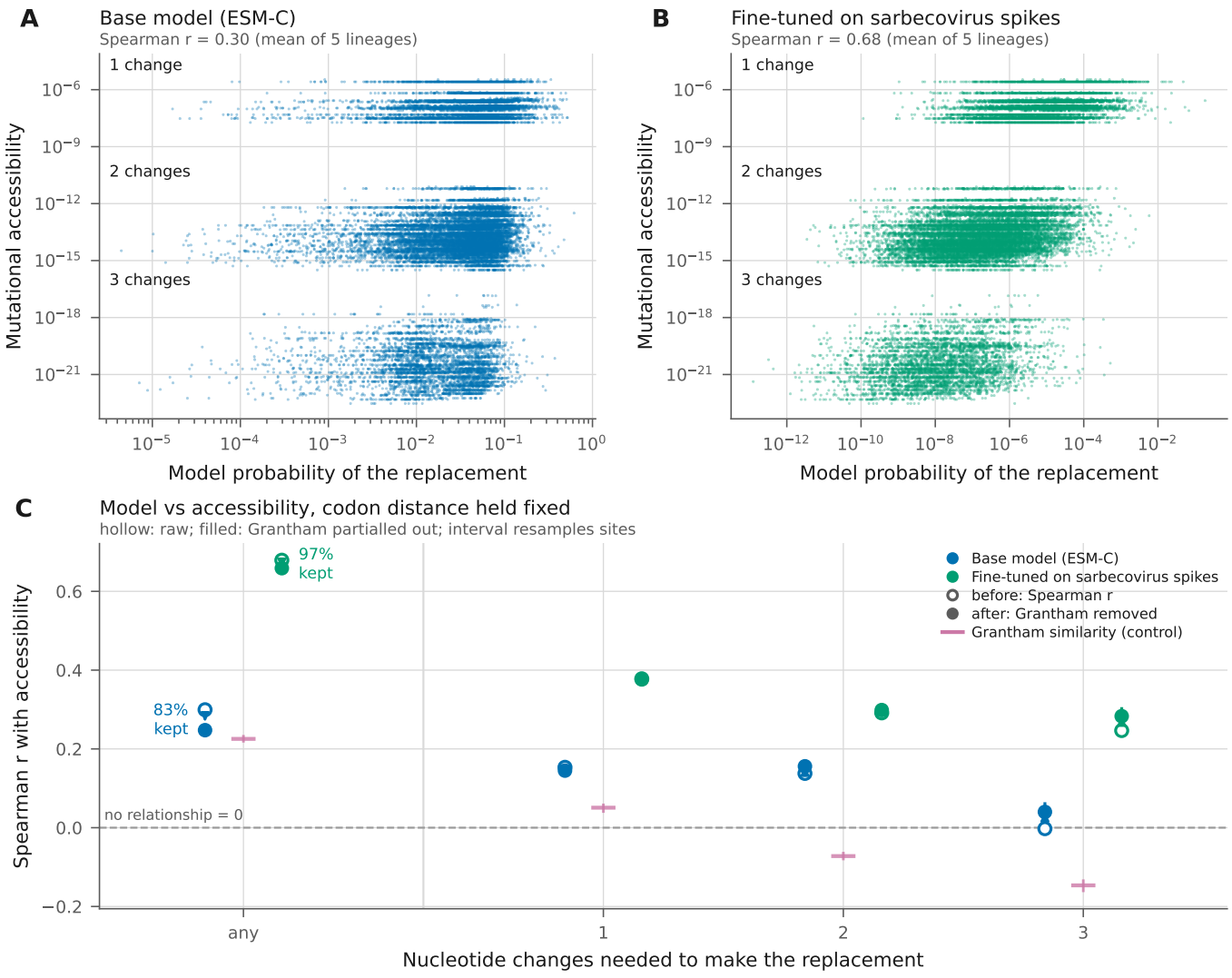

A
Base model (ESM-C)
Spearman r = 0.30 (mean of 5 lineages)
1 change
2 changes
3 changes
Mutational accessibility
Model probability of the replacement
B
Fine-tuned on sarbecovirus spikes
Spearman r = 0.68 (mean of 5 lineages)
1 change
2 changes
3 changes
Mutational accessibility
Model probability of the replacement
C
Model vs accessibility, codon distance held fixed
hollow: raw; filled: Grantham partialled out; interval resamples sites
97%
kept
83%
kept
no relationship = 0
Base model (ESM-C)
Fine-tuned on sarbecovirus spikes
before: Spearman r
after: Grantham removed
Grantham similarity (control)
Spearman r with accessibility
any
1
2
3
Nucleotide changes needed to make the replacement